BEAM DESIGN CALCULATION

Project Name	:	G+2 RB BEAM DESIGN @SMR
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-13.625 m.R1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 00:54:36
Level Designed	:	13.625 m

Sr.No.	Symbol		Definitions
1	α	=	Angle formed with horizontal by diagonal reinforcement
2	ε1	=	Strain at level considered, Calculated ignoring the stiffening of the concrete in tension zone
3	εm	=	Average steel strain at level considered
4	Acr	=	Distance from the point considered to the surface of the nearest longitudinal bar
5	Adr	=	Area of diagonal reinforcement in coupler beam
6	AstCalc	=	Area of Tension reinforcement required in sqmm
7	Ast comp	=	Ast at compression side
8	AstCrack	=	Area of Tension reinforcement for Crack Width required in sqmm
9	AstPrv	=	Area of reinforcement provided at given section in sqmm
10	Asv	=	Area of shear reinforcement required per meter length in sqmm
11	Asv Reqd	=	Min area shear reinforcement as per clause 41.4.3
12	Asv Total Prv	=	Total area of shear reinforcement provided, including that for torsion
13	Asv Torsion	=	Area of Stirrups over the corner bars to resist shear and torsion as per clause 41.4.3
14	Asv Torsion Prv	=	Area of shear reinforcement resisting torsion (outer legs only)
15	b	=	Width of the Beam in mm
16	bw	=	Width of Web in mm
17	Ceff	=	Effective Cover of Beam in mm
18	Cmin	=	Clear cover to top and bottom reinforcement in mm
19	Cside	=	Clear cover to side reinforcement in mm
20	d	=	Effective depth of Beam in mm
21	d'	=	Clear cover to reinforcement at compression side.
22	D	=	Depth of Beam in mm
23	D'	=	Distance (c/c) between top-most and bottom-most layer
24	Df	=	Thickness of Flange in mm
25	Ec	=	Modulus of elasticity of concrete
26	Es	=	Modulus of elasticity of steel
27	FcPerm	=	Permissible Stress in Concrete required in N/sqmm

28	Fst	=	Stress in steel
29	FstPerm	=	Permissible Stress in Steel required in N/sqmm
30	Legs	=	Number of legs of the shear reinforcement
31	lo	=	Effective Length of Beam in mm
32	Mh	=	Hogging Moment of Resistance in kNm
33	Ms	=	Sagging Moment of Resistance in kNm
34	M_{Tu}	=	Converted Bending Moment due to Torsion at a section in kNm
35	Mu	=	Factored Bending Moment in Limit State at a section in kNm
36	Mud	=	Total Design Moment = Mu + M_{Tu} in kNm
37	MuLim	=	Limiting Moment of Resistence of Singly Reinforced Section
38	NA angle (φ)	=	It is a neutral axis angle corresponding to load angle to find out MCap
39	Pcclc	=	Calculated percentage compression steel
40	Ptclc	=	Calculated percentage tension steel
41	Ptmin	=	Minimum percentage steel as per code
42	PtPrv	=	provided percentage steel
43	R	=	Moment Ratio Mud/Bd^2
44	Reinf	=	Rebar arrangement provided
45	sp	=	Spacing Between bars at outer most layer
46	Spc1 to Spc6	=	Max Stirrup Criteria in mm
47	Stirrup Dia	=	Diameter of shear stirrups in mm
48	SvCalc	=	Stirrup spacing calculated as per Asv in mm
49	SvPrv	=	Stirrup spacing provided in m
50	Tc	=	Permissible Shear stress in concrete at a given section in N/sqmm Calculated as per Tensile reinforcement
51	Tce	=	Earthquake inducted shear stress in N/sqmm
52	Tcep	=	Permissible value of shear stress for checking of coupling action in N/sqmm
53	Tu	=	Factored Torsional Moment in Limit State at a section in kNm
54	Tv	=	Shear stress due to Vud in N/sqmm
55	V^{D+L}	=	Shear force 1.2x(Dead Load + Live Load) in kN
56	Vc	=	Shear capacity of concrete section due to Tc in kN
57	Vu	=	Factored Shear Force in Limit State at a section in kN
58	Vu-A1	=	V^{D+L}Left - 1.4 x (MsLeft + MhRight)/L in kN
59	Vu-A2	=	V^{D+L}Left + 1.4 x (MhLeft + MsRight)/L in kN
60	Vu-B1	=	V^{D+L}Right + 1.4 x (MsLeft + MhRight)/L in kN
61	Vu-B2	=	V^{D+L}Right - 1.4 x (MhLeft + MsRight)/L in kN
62	Vud	=	Design Shear Force (Max of Vus and Vu-sway) in kN
63	Vue	=	Earthquake inducted shear in kN
64	Vu-sway	=	Max of (Vu-A1 and Vu-A2) & (Vu-B1 and Vu-B2) in kN
65	Vus	=	Vud - Vc Design Shear Force for reinforcement, in kN
66	Vut	=	Total Shear Force = Vu + V_{Tu} in kN

67	V_{Tu}	=	Converted Shear Force due to Torsion at a section in kN
68	Wcr	=	Surface Crack Width
69	WcrPerm	=	Permissible Crack Width required in mm
70	X1	=	Smaller c/c dimension of outermost Stirrups
71	Y1	=	Larger c/c dimension of outermost Stirrups
			All 6mm diameter bars are assumed 250 grade with fy = 250 N/sqmm

All Forces are in 'kN', 'kNm', Stress in 'N/sqmm' & Dimension are in 'mm'.

Code References

IS 456

Sr.No.	Item		Clause / Table
1.	Ptmax	:	26.5.1.1
2.	Ptmin	:	26.5.1.1
3.	M_{Tu}	:	41.4.2
4.	V_{Tu}	:	41.3.1
5.	Ptclc	:	38
6.	Tc	:	40.2.1
7.	Tcmax	:	40.2.3
8.	Asv	:	40.4
9.	Min Shear Reinf	:	26.5.1.6
10.	Max Stirrup Spacing	:	26.5.1.5
11.	Shear Reinf - Torsion	:	26.5.1.7
12.	Side Face Reinforcement	:	26.5.1.3
13.	crack width calculation	:	Annex-F
14.	cracking	:	Clause 35.3.2

IS 13920 - 2016

Sr.No.	Item		Clause / Table
1.	Ptmax	:	6.2.2
2.	Ptmin	:	6.2.1
3.	Astmin	:	6.2.3 & 6.2.4
4.	Svclc	:	6.3.5
5.	Shear Capacity	:	6.3.4
6.	Coupling Beams	:	10.5

Group	:	**G1**
Beam No	:	**B1**
Analysis Reference (Member)	:	177 (13.625 m)
Beam Length	:	3924 mm

Breadth (B)	:	230	mm
Depth (D)	:	300	mm
Effective Depth (d)	:	260	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	12	5	14	14	17	12
Critical L/C - RCDC	8	1	10	10	13	8
Mu (kNm)	6.31	7.27	6.19	11.36	2.27	14.13
Tu (kNm)	0	0.17	0	0	0	0
M_{Tu} (kNm)	0	0.23	0	0	0	0
Mud (kNm)	6.31	7.5	6.19	11.36	2.27	14.13
MuLim (kNm)	51	51	51	51	51	51
R	0.406	0.482	0.398	0.731	0.146	0.909
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0	0
PtPrv (%)	0.378	0.378	0.378	0.378	0.378	0.378
AstCalc (sqmm)	119.6	119.6	119.6	119.6	119.6	119.6
AstPrv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement Provided	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required

D = AstCrack = Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	14	12	12
Critical L/C - RCDC	10	8	8
PtPrv (%)	0.378	0.378	0.378
Vu (kN)	13.19	8.59	14.17
Tu (kNm)	0	0	0
V_{Tu} (kN)	0	0	0
Vut (kN)	13.19	8.59	14.17
Tv (N/sqmm)	0.22	0.14	0.24
Tc (N/sqmm)	0.44	0.44	0.44
Vc (kN)	26.04	26.04	26.04
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	-	-
Asv Torsion Prv (sqmm)	-	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	195	195	195
SvPrv (mm)	195	195	195
Asv Total Prv (sqmm)	515.59	515.59	515.59

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 195	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 300	mm
Web Depth	= 300 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G2**
Beam No	:	**B2**
Analysis Reference (Member)	:	179 (13.625 m)

Beam Length	:	3924	mm
Breadth (B)	:	230	mm
Depth (D)	:	300	mm
Effective Depth (d)	:	260	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	12	5	14	14	5	12
Critical L/C - RCDC	8	1	10	10	1	8
Mu (kNm)	4.61	7.83	5.61	9.52	-	8.62
Tu (kNm)	0.3	0.44	0.19	0.19	0.44	0.3
M_{Tu} (kNm)	0.4	0.6	0.25	0.25	0.6	0.4
Mud (kNm)	5.01	8.43	5.87	9.77	0.6	9.02
MuLim (kNm)	51	51	51	51	51	51
R	0.322	0.542	0.377	0.628	0.039	0.58
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0	0.008	0
PtPrv (%)	0.378	0.378	0.378	0.378	0.378	0.378
AstCalc (sqmm)	119.6	119.6	119.6	119.6	119.6	119.6
AstPrv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement Provided	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A	=	Ptmin	=	Min % area of flexural reinforcement
B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement

C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	5	14	12
Critical L/C - RCDC	1	10	8
PtPrv (%)	0.378	0.378	0.378
Vu (kN)	12.57	6.78	11.72
Tu (kNm)	0.44	0.19	0.3
V_{Tu} (kN)	3.08	1.31	2.06
Vut (kN)	15.65	8.09	13.78
Tv (N/sqmm)	0.26	0.14	0.23
Tc (N/sqmm)	0.44	0.44	0.44
Vc (kN)	26.04	26.04	26.04
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	95.104	47.395	79.723
Asv Torsion Prv (sqmm)	515.59	515.59	515.59
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	195	195	195
SvPrv (mm)	195	195	195
Asv Total Prv (sqmm)	515.59	515.59	515.59

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 195	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 300	mm
Web Depth	= 300 <= 750	mm

Side Face Reinforcement not required.

Group : **G3**

Beam No : **B3**

Analysis Reference (Member)	:	180 (13.625 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	300	mm
Effective Depth (d)	:	260	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	5	5	18	9	15	15
Critical L/C - RCDC	1	1	14	5	11	11
Mu (kNm)	8.77	6.83	1.32	-	3.17	9.57
Tu (kNm)	0.54	0.54	0.27	0.66	0.34	0.34
M_{Tu} (kNm)	0.74	0.74	0.36	0.89	0.46	0.46
Mud (kNm)	9.5	7.57	1.68	0.89	3.63	10.03
MuLim (kNm)	51	51	51	51	51	51
R	0.611	0.487	0.108	0.057	0.233	0.645
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.2	0.2	0.2	0.2	0.2	0.2
Pcclc (%) (C)	0	0	0	0.012	0	0
PtPrv (%)	0.378	0.378	0.378	0.378	0.378	0.378
AstCalc (sqmm)	119.6	119.6	119.6	119.6	119.6	119.6
AstPrv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement Provided	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)

AstCalc = Bn x B X d (for Mu = 0)

Where,

A = Ptmin = Min % area of flexural reinforcement

B' = Ptclc = Maximum of % reinforcement required for Bending Moment and Min % area of flexural

				reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design			
	Left	**Mid**	**Right**
Critical L/C - Analysis	15	15	5
Critical L/C - RCDC	11	11	1
PtPrv (%)	0.378	0.378	0.378
Vu (kN)	6.14	10.82	12.22
Tu (kNm)	0.34	0.34	0.54
V_{Tu} (kN)	2.35	2.35	3.78
Vut (kN)	8.49	13.17	16
Tv (N/sqmm)	0.14	0.22	0.27
Tc (N/sqmm)	0.44	0.44	0.44
Vc (kN)	26.04	26.04	26.04
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	55.792	78.556	101.146
Asv Torsion Prv (sqmm)	515.59	515.59	515.59
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	195	195	195
SvPrv (mm)	195	195	195
Asv Total Prv (sqmm)	515.59	515.59	515.59

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 195	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 300	mm
Web Depth	= 300 <= 750	mm

Side Face Reinforcement not required.

Group	:	**G4**	
Beam No	:	**B4**	
Analysis Reference (Member)	:	178 (13.625 m)	
Beam Length	:	2134	mm
Breadth (B)	:	230	mm
Depth (D)	:	300	mm
Effective Depth (d)	:	260	mm
Design Code	:	IS 456	
Beam Type	:	Regular Beam	
Grade Of Concrete (Fck)	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover (Cmin)	:	30	mm
Side Clear Cover	:	30	mm
Es	:	2x10^5	N/sqmm
Ptnominal (%) (Bn)	:	0.2	%

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Critical L/C - Analysis	5	5	12	12	15	5
Critical L/C - RCDC	1	1	8	8	11	1
Mu (kNm)	16.1	10.03	-	-	6.4	16.95
Tu (kNm)	0	0	0.59	0.59	0.19	0
M_{Tu} (kNm)	0	0	0.8	0.8	0.25	0
Mud (kNm)	16.1	10.03	0.8	0.8	6.66	16.95
MuLim (kNm)	51	51	51	51	51	51
R	1.035	0.645	0.051	0.051	0.428	1.09
Ptmin (%) (A)	0.2	0.2	0.2	0.2	0.2	0.2
Ptclc (%) (B')	0.228	0.2	0.2	0.2	0.2	0.241
Pcclc (%) (C)	0	0	0.011	0.011	0	0
PtPrv (%)	0.378	0.378	0.378	0.378	0.378	0.378
AstCalc (sqmm)	136.27	119.6	119.6	119.6	119.6	143.91
AstPrv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement Provided	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Note: Calculation of Ast

AstCalc = Max {B',C} x B X d (for Mu > 0)
AstCalc = Bn x B X d (for Mu = 0)
Where,
A = Ptmin = Min % area of flexural reinforcement

B'	=	Ptclc	=	Maximum of % reinforcement required for Bending Moment and Min % area of flexural reinforcement (A)
Bn	=	Ptnominal	=	Nominal % area of reinforcement
C	=	Pcclc	=	% area of compression reinforcement required
D	=	AstCrack	=	Area of reinforcement required for Crackwidth

Shear Design

	Left	Mid	Right
Critical L/C - Analysis	5	5	5
Critical L/C - RCDC	1	1	1
PtPrv (%)	0.378	0.378	0.378
Vu (kN)	12.76	18.23	19.98
Tu (kNm)	0	0	0
V_{Tu} (kN)	0	0	0
Vut (kN)	12.76	18.22	19.98
Tv (N/sqmm)	0.21	0.3	0.33
Tc (N/sqmm)	0.44	0.44	0.44
Vc (kN)	26.04	26.04	26.04
Vus=Vut-Vc (kN)	0	0	0
Legs	2	2	2
Stirrup Rebar	8	8	8
Asv Torsion (sqmm)	-	-	-
Asv Torsion Prv (sqmm)	-	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
SvCalc (mm)	195	195	195
SvPrv (mm)	195	195	195
Asv Total Prv (sqmm)	515.59	515.59	515.59

Maximum Spacing Criteria

Basic

Spc1 =0.75d	= 195	mm
Spc2	= 300	mm

SFR Design

Beam Width	= 230	mm
Beam Depth	= 300	mm
Web Depth	= 300 <= 750	mm

Side Face Reinforcement not required.